

SemantikML UI Manual Test Cases

Manual browser-based test cases for the Streamlit UI, intended for end-user validation rather than automated tests.

Core User Flows

- **1. Landing page loads**
Open the app and verify the hero section, domain selector, search method selector, top-k slider, query field, and Start Search / Reset Search buttons are visible.
- **2. Empty search**
Leave the query blank and click Start Search. Expected: no crash, no results, no answer, and no misleading error.
- **3. Reset search**
Fill all fields, click Reset Search. Expected: query clears, defaults return, current page stays stable, and login state is preserved if already signed in.
- **4. Domain switching**
Run the same query once each in Research, Blog, and Documentation. Expected: results reflect the selected domain and the chosen domain is shown on result cards.
- **5. Top-k bounds**
Test with Top-K = 1 and Top-K = 20. Expected: UI remains responsive and result count behaves sensibly.

Anonymous User Limits

- **1. Free-search counter visibility**
As a signed-out user, verify the sidebar shows remaining free searches.
- **2. First three searches**
Perform 3 valid searches while signed out. Expected: all 3 succeed, and the remaining counter decreases correctly.
- **3. Fourth search blocked**
Attempt a 4th valid search while signed out. Expected: redirect to login page with the message about using 3 free searches.
- **4. Empty query does not consume quota**
Signed out, click Start Search with an empty query. Expected: free-search counter does not decrease.

Signup and Login

- **1. Signup validation**
Try creating an account with username under 3 chars, invalid email, password under 8 chars, and mismatched confirmation. Expected: each invalid input shows a clear error.
- **2. Successful signup**
Register with a valid new username/email/password. Expected: success message and redirect to login.
- **3. Duplicate signup**
Try signing up again with the same username, then with the same email. Expected: specific duplicate error messages.
- **4. Login by username**
Sign in with username + password. Expected: redirect to landing page and authenticated sidebar appears.

- **5. Login by email**

Sign in with email + password. Expected: login works exactly the same.

- **6. Invalid login**

Try wrong password and unknown user. Expected: Invalid username or password.

- **7. Case-insensitive email login**

If your account email is lowercase, try logging in with uppercase/mixed-case email. Expected: login still works.

Search Results

- **1. Semantic search results**

Use Semantic Search with a realistic query. Expected: results page shows result cards with title, domain, source, score, and snippet.

- **2. RAG answer mode**

Use RAG Answer with a realistic question. Expected: answer box appears, related papers appear, and a Thought time value is shown.

- **3. Switching methods**

Search once with Semantic Search, then switch to RAG Answer for the same query. Expected: the page updates cleanly and does not show stale content from the previous mode.

- **4. Long query handling**

Paste a long natural-language query. Expected: UI remains usable and returns either results or a graceful empty state.

Similar Documents

- **1. Find similar from results**

On a result card, click Find Similar. Expected: navigation to Similar Documents page.

- **2. Similar-results content**

Verify each similar result shows title, domain, source, score, and snippet.

- **3. No-similar-results path**

Try a narrow or unusual query, then Find Similar. Expected: if nothing qualifies, the page shows the no-similar-documents message.

- **4. Back navigation**

On the similar page, click Back to Search. Expected: returns to search page without losing the original result set.

- **5. Home logo navigation**

From search or similar page, click the SemantikML logo/home control. Expected: returns to landing page.

Authenticated History

- **1. History visibility**

After logging in, verify the sidebar shows Search History instead of the anonymous free-search counter.

- **2. Save search to history**

Run a search while logged in. Expected: the query appears in history.

- **3. Re-run from history**

Select a previous search in the history dropdown. Expected: query, domain, method, and top-k are restored, and the search reruns automatically.

- **4. Duplicate query deduplication**

Run the exact same query twice while logged in. Expected: only one history entry remains for that query, updated to the latest settings/time.

- **5. Case-insensitive deduplication**

Run transformers and then Transformers. Expected: still one history entry rather than two.

- **6. Clear history**

Click Clear history. Expected: history list becomes empty and the sidebar shows no saved searches yet.

- **7. History is user-specific**

If you can use two accounts, confirm searches from account A do not appear in account B.

Logout

- **1. Logout flow**

Log in, then click Log Out. Expected: app returns to landing, authenticated UI disappears, and anonymous UI returns.

- **2. Post-logout state reset**

After logout, verify the free-search counter is reset and no private history remains visible.

Admin-Only Checks

- **1. Non-admin user**

Log in with a normal user. Expected: no Admin Panel button in the sidebar.

- **2. Admin user**

Log in with an admin email. Expected: Admin Panel button is visible.

- **3. Admin users tab**

Open admin panel and verify the Users tab lists registered users.

- **4. Admin search activity tab**

Run a few searches, then check Search Activity. Expected: recent searches appear with query, domain, method, top-k, and result count.

- **5. Admin auth log tab**

Perform signup, failed login, successful login, and logout, then verify those events appear in Auth Log.